$$5x_{i-1} - 26x_i + 5x_{i+1} = 0$$

$$x_0 = 1$$

$$x_1 = \frac{1}{5}$$

$$x_{i-1} := x$$

$$5x^2 - 26x + 5 = 0$$

$$\Delta = 676 - 100 = 576$$

$$\sqrt{\Delta} = 24$$

$$x_1 = \frac{26-24}{10} = \frac{1}{5}$$

$$x_2 = \frac{26+24}{10} = 5$$

$$A\left(\frac{1}{5}\right)^n + B(5^n) = x_n$$

$$\begin{cases} x_0 = 1 \\ x_1 = \frac{1}{5} \end{cases} \Rightarrow \begin{cases} A + B = 1 \\ \frac{1}{5}A + 5B = \frac{1}{5} \end{cases} \Rightarrow \begin{cases} B = 1 - A \\ \frac{1}{5}A + 5 - 5A = \frac{1}{5} \end{cases} \Rightarrow \begin{cases} B = 1 - A \\ A = 1 \end{cases} \Rightarrow \begin{cases} B = 0 \\ A = 1 \end{cases}$$

zatem $x_n = \left(\frac{1}{5}\right)^n = \frac{1}{5^n}$

więc $\frac{1}{5^n}$ jest jedynym ciągiem spełniającym warunki